MATH 20250 HW 1 SUMMARY: MISTAKES, MISCONCEPTIONS, AND HINTS

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1. Q1.1.7

The question asks us to prove $0v = 0$.

A Wrong Proof. By distribution, we have for any vector v and scalar a:

$$av + (-a)v = 0v$$

By associativity, we have

$$0 = av + (-av) = 0v.$$

$\square$

The proof is wrong in its last step $(-a)v = -av$. Saying "by associativity," the student here probably wants to say $((-1)a)v = -1(av)$, but this requires Q1.1.8. Unless Mr. B has a way of proving Q1.1.8 without Q1.1.7, this would result in circular reasoning.

In some cases, the mistake is covered by the student's use of subtraction as a notation.

The Same Wrong Proof Disguised. By distribution, we have for any vector v and scalar a:

$$0v = (a - a)v = av - av = 0$$

$\square$

In writing $(a - a)v = av - av$, the student is actually saying $(a + (-a))v = av + (-av)$, i.e. $(-a)v = -av$, which requires Q1.1.8.

The root of both types of mistakes is eventually that the students have not get used to dealing with vector spaces abstractly, and is prone to bringing their intuition in $\mathbb{R}$ into the algebra of vectors. (Utilizing one's intuition is not a bad

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thing, yet intuition should be used with care, lest the student make mistakes like those presented here.)

> The point of rigour is not to destroy all intuition; instead, it should be used to destroy bad intuition while clarifying and elevating good intuition. —Terence Tao

The antidote to the problem, then, is to stick to the axioms and definitions, especially while we are still getting familiar with the structure in LADW 1.1. Specifically, we should be cautious that subtraction $a - b$, which we are familiar with, is not inherent in the definition of vector space, but short-hand for adding the additional inverse $a + (-b)$. This caution is going to be helpful if the students is to continue their study in the Basic Algebra sequence.

2. Q1.1.8

The question asks us to prove $(-1)v = -v$. In answering this question, some students tend to assume the space is isomorphic to $\mathbb{R}^n$, and then conflate the idea vector with the column vectors.

Part of the cause for this mistake is probably also that the student is not comfortable with considering vector spaces abstractly.

3. Q1.2.6

Let $v_1 = av_2 + bv_3$. Define

$$w_1 = v_1 + v_2, w_2 = v_2 + v_3, w_3 = v_3 + v_1$$

We are required to prove that w_1, w_2, w_3 are linear dependent. This is done by constructing w_2 as linear combination of w_1 and w_3.

In at least two student responses, the coefficients of w_1 and w_3 is would contain a fraction where something like $a + b + 1$ is in the denominator. In this case, the students often fail to consider the case when $a+b+1 = 0$ (or skip the consideration because they think this case is trivial).

A Proof that would Get Stuck, Written by Myself. Because $v_1 = av_2+bv_3$, we have

$$w_1 = (a + 1)v_2 + bv_3$$
$$w_3 = av_2 + (b + 1)v_3.$$

Thus,

$$w_1 + w_3 = (a + b + 1)w_2$$

□

We are then tempted to write $w_2 = \frac{w_1+w_3}{a+b+1}$, but this make sense only when $a + b + 1 \neq 0$. We shall solve this problem by considering the cases when $a + b + 1$ is zero or non-zero.

The Proof Unstuck. If $a + b + 1 = 0$, then $w_1 + w_3 = 0$, meaning the two are linear dependent, so w_1, w_2, w_3 are linear dependent.

If $a + b + 1 = 0$, then $w_2 = \frac{w_1+w_3}{a+b+1}$, so w_1, w_2, w_3 are linear dependent. □

4. Q1.3.1(d)

At least two students tried to give an answer to the multiplication, when the multiplication is undefined.

5. Q1.3.3(c-d)

The question asks about expressing derivation and second-order derivation as matrices. This requires treating P_n the space of polynomials with at most order n as a vector space, and selecting a basis for the space P_n, which is already given as $1, t, ..., t^n$ in that order.

(Notice that there are $n+1$ vectors in this basis, so the space is $n+1$-dimensional, so a polynomial written as a column vector is supposed to have $n+1$ entries, so our desired matrix would be $n+1 \times n+1$. In some cases, the student's response is not even a square matrix.)

Given a basis for P_n, we can write a polynomial of at most order n as a column vector, and here is the thing: when we write $f = a_0 + ... + a_n t^n$ as a column vector: **the entries of the column vector should be the coefficients of the polynomial, not the monomial components of the polynomial**.

Why is this? The reason is that when we write a vector v as a column vector, the entries are supposed to be the scalar coefficients, when v is written as a unique linear combination. (See LADW 1.2.)

Thus, since we have selected basis $1, t, ..., t^n$, f as a column vector should be as such:

$$\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \\ a_n \end{pmatrix}.$$

Similarly, the derivative $f' = a_1 + 2a_2 t + ... + na_n t^{n-1} + 0t^n$ should be

$$\begin{pmatrix} a_1 \\ 2a_2 \\ \vdots \\ na_{n-1} \\ 0 \end{pmatrix}$$

Knowing this, and with enough caution, the students are able to figure out the matrix on themselves. If the students mistakenly believe that the entries are monomial components of the polynomials, wrong answers would naturally follow.

Example of Such an Expression, by Mr. B. When $f = a_0 + ... + a_n t^n$, we have

$$f = \begin{pmatrix} a_0 \\ a_1 t \\ \vdots \\ a_{n-1} t^{n-1} \\ a_n t_n \end{pmatrix}.$$

Taking derivative, we have:

$$f' = \begin{pmatrix} 0 \\ a_1 \\ \vdots \\ (n-1)a_{n-1} t^{n-2} \\ na_n t_{n-1} \end{pmatrix}.$$

□

6. Supplementary Problem 1

For the number of bases of $(\mathbb{Z}/5\mathbb{Z})^2$, consider the procedure: we choose two non-zero vectors from the space. For the first vector we have 24 choices (all except for 0), and for the second vector we have 20 choices (vectors that are linear independent with the first vector). Thus there are 480 choices in total. This is divided by the permutation number which is 2. We yield 240 as the number of bases.

We can similarly compute the number of bases for $(\mathbb{Z}/p\mathbb{Z})^n$: for the first vector v_1 we have $p^n - 1$ possibilities since we need only $v_1 \neq 0$. For the second vector v_2 we require $v_2 \notin Span(v_1)$ so there are $p^n - p$ possibilities. For each subsequent v_{i+1}, we require $v_{i+1} \notin Span(v_1, ..., v_i)$, so there are $p^n - p^i$ possibilities. We then divide the entire product by $n!$ since there are $n!$ kinds of permutation of n distinct vectors. Thus the result would be

$$\frac{1}{n!}\prod_{i=0}^{n-1}(p^n - p^i).$$